Real-Time Smoking Detection using YOLOv11 and Mediapipe Holistic Fusion

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***Abstract*—Public smoking is a significant health and regulatory concern that requires the need for automated surveillance systems capable of effectively identifying smoking behaviors in real-world scenarios. Conventional object detection models such as YOLO have been observed to face difficulties in detecting cigarette objects due to their small size, occlusion, and similarity to other objects, thereby leading to missed detections. This paper proposes a multimodal smoking detection system that integrates YOLOv11 object detection with Mediapipe Holistic landmark analysis to tap into the benefits of object detection and human behavior.A rule-based late fusion approach is used to validate YOLO detections based on contextual human landmarks (hands, pose, and face) to filter out false positives and enhance behavioral consistency.The model is trained on the Roboflow Smoking Person Detection dataset, which contains 2,349 images, and the results are tested using the pretrained YOLOv11 weights. The experimental results show that the proposed fusion approach has an F1-score of 0.805 with a precision of 0.97, which is higher than the YOLOv11-only approach (0.727) and YOLOv8-based approach (0.7749). To address the limitation of frame-level prediction, a sliding window temporal smoothing technique was proposed, which has increased the accuracy to 0.8229 and F1-score to 0.9006 with a substantial reduction in false negatives. The runtime performance analysis shows that the average processing rate is approximately 8 FPS, which is adequate for continuous monitoring.An ablation study has also been performed to demonstrate the efficacy of the designed rule-based fusion strategy in improving recall without compromising precision. The proposed system provides a reliable, interpretable, and privacy-respecting solution for automated smoking detection on offline image and video datasets, indicating its appropriateness for future surveillance applications.**

***Index Terms*—YOLOv11, Mediapipe Holistic, Smoking Detection, Multimodal Fusion, Human Pose Estimation, Temporal Smoothing, Computer Vision, Behavioral Activity Recognition, Public Surveillance, Deep Learning.**

I. Introduction

Public smoking is a serious source of health hazards, environmental risks, and legal violations. Exposure to second-hand smoke is one of the main causes of preventable diseases such as respiratory and cardiovascular ailments, while cigarette butts are a major source of environmental pollution. Because of these factors, public organizations such as universities, hospitals, and transport hubs have a strict no-smoking policy. The monitoring system must therefore be efficient in such environments. Manual CCTV monitoring is also extremely impractical for large-scale surveillance as it is time-consuming, error-prone, and incapable of providing consistent real-time feedback. These factors make the need for automated smoking detection systems that can facilitate efficient enforcement of no-smoking policies even more pressing. Deep learning-based vision systems have been employed in the past few years for automating the detection of prohibited behaviors. The YOLO (You Only Look Once) series of models has demonstrated extremely robust performance in general object detection applications because of its ability to perform real-time inference with high accuracy [1], [2]. Cigarette detection is a particularly challenging task despite such success stories. Cigarettes are usually extremely small compared to the human body, often partially occluded by fingers or during hand-to-mouth actions. Moreover, their rod-like structure and low reflectance make them highly confusable with harmless objects such as pens, lighters, or straws [3]. The presence of all these factors usually

leads to extremely high false negative rates and low recall when tested on real-world CCTV footage, especially when the presence of variable lighting, motion blur, and low resolution is considered [4].To address these issues, alternative solutions have been sought by researchers. Multimodal models that integrate object detection with additional secondary context information have been designed in some research works to improve detection accuracy [5]. Other researchers have also explored the use of wearable sensor-based approaches, such as wrist-worn accelerometers and gyroscopes, to detect smoking behaviors based on motion patterns [6]. While these approaches have shown promise, they are typically constrained by domain specificity, requiring user compliance with wearable devices, or lack scalability in public surveillance scenarios. Other research has sought to improve the detectors themselves by enhancing feature extraction modules, adding attention mechanisms, or optimizing small-object detection [7]. While these improvements have increased accuracy on controlled datasets, they remain challenged by unconstrained real-world scenarios. These observations underscore the gap between theoretical prototype development and practical CCTV-ready system development. In practice, a system requires not only strong detection performance but also extremely low false alarm rates, since excessive false alarms erode operator confidence and increase monitoring tasks. Motivated by the gap, this paper introduces a YOLOv11 + Mediapipe Holistic fusion system. Unlike the traditional YOLO-alone architectures, our proposed system validates cigarette detections based on contextual hand, face, and body keypoints to ensure behavioral consistency. The fusion strategy improves robustness by removing false alarms while maintaining high detection accuracy, and the developed system can be applied to a range of surveillance scenarios.

II. Related Works

Current studies on smoking detection have been categorized into three categories: object detection–only approaches, fusion with pose or landmark-based contextual cues, and multimodal sensing approaches. The initial category employed object detection–only models [8]. [8] contrasted YOLOv5, EfficientDet-D2, and EasyDL with AutoML and recorded 81–84% mAP. Following this, [9] presented Smoking-YOLOv8 with attention modules and recorded mAP@0.5 $\approx$ 83.7% in industrial environments. Likewise, [10] experimented with YOLOv8 with different backgrounds and recorded medium accuracy but observed difficulty in detecting small cigarettes. To enhance robustness, [4] introduced DAHD-YOLO, and [11] introduced YOLOv8-MNC; both enhanced small target detection but continued to be plagued by occlusion and false positives. Another recent effort by [12] focused on fire exit areas, where their self-designed CCTV model reported recall = 78.90% and mAP@0.5 = 83.70%, both performing better than the YOLOv8/v11/v12 baselines but still constrained in low-light environments. The second category merged object detection with pose or landmark-based contextual cues. [13] used YOLOv5 integrated with MediaPipe landmarks, achieving 92.7% accuracy for smoking detection in a classroom environment. [14] merged YOLO, AlphaPose, and ST-GCN, registering F1-scores of 78–82% by encoding temporal hand-to-mouth gestures, although the process was computationally intensive. The third path investigated multimodal sensing methods. [15] employed wrist-worn inertial sensors to detect puffs with F1 = 0.863 in natural environments. [16] surveyed deep learning models for classifying smokers and verified that medium accuracy is still prevalent under unconstrained conditions. [17] introduced a multimodal architecture with constrained training data, reaching 74% accuracy under image-only operation, showing promise but requiring larger datasets. In summary, object detection–only approaches [8], [17] typically achieve 81–84% mAP, gesture-fusion approaches [13], [14] are more accurate but computationally expensive, and multimodal systems [15], [16] are valuable but not scalable. Our YOLOv11 + MediaPipe Holistic fusion model reached F1 = 0.805 with precision = 0.97, outperforming YOLO-only baselines while reducing false positives, making it safer for deployment in real-world CCTV environments.

III. Methodology

A. Overview

The proposed approach employs a dual-branch architecture that integrates object detection and human landmark analysis to identify smoking activity in real-world surveillance videos. The dual-branch architecture involves the simultaneous processing of image/video data through two branches: YOLOv11 for the detection of people and cigarettes, and the Mediapipe Holistic approach for the detection of body, hand, and face landmarks. The fusion module then integrates the outputs of the two branches using rule-based validation to determine smoking activity. The proposed approach provides binary classification (smoking vs. non-smoking) with visual representations such as bounding boxes and landmarks. The proposed approach enhances the detection of small and concealed cigarettes by exploiting behavioral context that may not be detected by conventional object detection approaches.

B. Input Layer

The model supports static image inputs (JPG, JPEG) and video stream inputs (MP4). In video stream input, frames are extracted and processed in parallel to both YOLOv11 and Mediapipe Holistic branches. The parallel structure allows for the simultaneous extraction of object-level features and behavioral cues, which helps in improving the robustness of the model in the CCTV setting where cigarettes are often blurred or hidden.

C. YOLOv11 Path

The YOLOv11 branch is an object detection model that focuses on detecting people and smoking objects in the security camera video streams. The model comprises four key components that include preprocessing, backbone, neck, and detection head and post-processing modules.

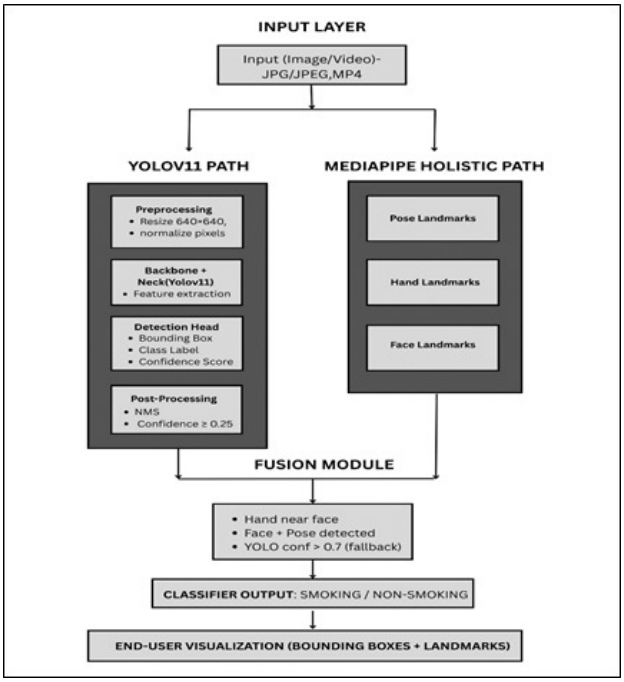

Fig. 1. Proposed YOLOv11 + Mediapipe Holistic Fusion Pipeline for Smoking Detection.

1) Preprocessing: The system resizes each video frame to 640×640 pixels before processing because YOLOv11 needs a specific input size of 640 pixels. Normalization aims to reduce variations that arise from differences in illumination levels and background noises common in CCTV surveillance systems. The preprocessing techniques create uniform input frames that help the model perform consistently in different settings.

2) Backbone and Neck: The YOLOv11 backbone network has functions that acquire both basic and advanced features from the input image. The features record basic visual elements that comprise edges and textures, as well as advanced object patterns. The neck module integrates the features using multi-scale representation, which improves the model's ability to detect small and concealed objects. The model must detect cigarettes effectively since, in surveillance videos, cigarettes appear in small sizes and unclear images, as well as in areas where they are concealed.

3) Detection Head: The detection head takes feature maps to generate output predictions that contain bounding boxes and class probabilities and confidence scores for all detected objects. The system works with YOLOv11, which has two detection classes set to detect persons and smoke. The detection system detects cigarette-like regions during this stage, which helps it to detect cigarettes even in environments with multiple people and background objects.

4) Post-Processing: The detection head generates raw predictions that contain overlapping bounding boxes. The system uses Non-Maximum Suppression (NMS) to eliminate redundant boxes while retaining the detection with the highest confidence. The system uses a confidence threshold of 0.25 to eliminate predictions that indicate low confidence levels. The detection stage is improved by these methods, but YOLOv11 still struggles to detect cigarettes that are obscured by hands or when they become blurred due to fast motion.

D. Mediapipe Holistic Path

The Mediapipe Holistic model enhances YOLOv11 by incorporating the capability to identify points on the human body that indicate the presence of surrounding environmental information. The main difference between the Mediapipe model and YOLO is in their object detection systems, where Mediapipe detects smoking behavior by recognizing gestures and postures, while YOLO detects objects only. The model detects body positions by using pose landmarks that recognize the movement of arms and body, hand landmarks that recognize contextual information about the presence of the hand near the detected object, and facial landmarks that recognize the position of the face. The model uses various indicators to detect smoking behavior, which is distinguished from normal actions that involve pen touching and face touching.

E. Fusion Module

The fusion module is based on a late decision-level rule-based fusion approach. If YOLOv11 has detected smoke with a confidence level above 0.25, the detection is then verified based on contextual information from Mediapipe Holistic. A detection is considered valid if hand landmarks are available, both pose and face landmarks are detected, which is an indicator of human activity context, or if the YOLO confidence level is above a fallback threshold of 0.7. This approach is effective in suppressing false positives due to similar objects while still being precise. The fusion is done at the decision level without calculating geometric hand-to-mouth distances.

IV. Results and Discussion

A. Baseline Performance of YOLOv11

The assessment began with the Roboflow Smoking Person Detection dataset, which utilized the YOLOv11-only model for the purpose of benchmarking. The model was able to attain an F1-score of 0.727 by demonstrating outstanding performance for person detection with an F1-score of 0.901, but the smoke detection only managed to attain 0.549. The assessment clearly indicates that the system is able to detect people in video surveillance effectively, but the detection of cigarettes is not possible due to their small size and pen-like structure, which tends to get hidden behind their hands and the movement of the cigarettes resulting in a blurred image in actual video surveillance. The system tended to generate a large number of false negatives while detecting using YOLOv11-only.

B. Proposed YOLOv11 + Mediapipe Holistic Fusion

The combination of YOLOv11 with Mediapipe Holistic landmarks in the hybrid system brought significant performance improvements compared to the baseline model. The fusion system achieved an accuracy score of 0.6771 along

Fig. 2. Example output of the proposed YOLOv11 + Mediapipe Holistic system, showing bounding boxes and human landmarks used to classify smoking activity.

with precision 0.9697, recall 0.6882, and F1-score 0.805. The system demonstrated excellent precision, which indicated that it resulted in very few false alarm instances. As shown in Fig. 2, the system correctly identified more actual instances of smoking compared to the original YOLOv11 model since the fusion system enhances the recall value from 0.507 (smoke-only YOLOv11 baseline) to 0.6882 while maintaining high precision. The result of the evaluation demonstrates that the combination of YOLOv11 object detection and Mediapipe Holistic gesture verification performs better.

Fig. 3. Qualitative result of the proposed YOLOv11 + Mediapipe Holistic fusion system showing cigarette detection, human landmarks, and final smoking classification.

The Fig. 4 shows the qualitative output of the proposed fusion system, where YOLOv11 locates the cigarette and Mediapipe Holistic offers contextual human landmarks. The final smoking decision is made by rule-based validation of object detection based on the presence of landmarks.

C. Temporal Modeling Impact Analysis

Smoking activity is inherently a temporal process and cannot be modeled using isolated frame-level predictions in all cases. Although the proposed YOLOv11-Mediapipe Holistic fusion framework provided high precision at the frame level, occasional failure in transient smoking detection resulted in missed smoking activity. To overcome this drawback, a sliding window temporal smoothing approach was incorporated. A temporal aggregation window of five consecutive frames was considered, where smoking activity was validated if at least three frames reported positive detection. This majority voting approach helps stabilize predictions against occlusion, motion blur, or partial visibility of cigarettes. The quantitative comparison between frame-level predictions and temporally smoothed predictions is summarized in Table I.

TABLE I
TEMPORAL SMOOTHING PERFORMANCE COMPARISON

Metric	Frame-Level	Temporal-Smoothed	Improvement
Accuracy	0.6771	0.8229	+14.6%
Precision	0.9697	0.9872	+1.75%
Recall	0.6882	0.8280	+13.98%
F1-score	0.8050	0.9006	+9.56%

The addition of temporal smoothing helps improve the robustness of detection. Although the precision of frame-level fusion is high, it has the problem of missed detection due to short-term inconsistencies. The addition of temporal aggregation helps improve the recall value from 0.6882 to 0.8280 and the F1-score from 0.8050 to 0.9006. It is important to note that the false negatives were decreased by 44.8%, which is an important indication that smoking detection is helped by short-term temporal reasoning.

D. Temporal Consistency Enhancement

To further analyze the temporal stability, the sliding window approach was tested under continuous video processing. Smoking detection was ensured if at least three out of five consecutive frames showed positive classification.

After applying temporal smoothing:

- Accuracy improved from 0.6771 to 0.8229
- Recall increased from 0.6882 to 0.8280
- F1-score improved from 0.8050 to 0.9006
- False negatives decreased from 29 to 16

This is equivalent to a 44.8% reduction in the number of missed smoking events. The above results clearly show that the proposed lightweight temporal aggregation method greatly improves robustness without resorting to computationally intensive sequence learning models like LSTMs or transformers.

E. Computational Performance Analysis

The proposed system was evaluated using recorded video sequences to analyze runtime performance under continuous processing conditions. Each frame underwent object detection, holistic landmark extraction, and rule-based fusion validation.

The observed runtime statistics are summarized below:

- Total Frames Processed: 369
- Total Processing Time: 46.22 seconds
- Average FPS: 7.98
- Average Latency per Frame: 0.1253 seconds
- Total YOLO Detections: 551

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========== PERFORMANCE REPORT ==========
Total Frames Processed : 455
Average YOLO Time/frame : 0.0297 sec
Average Mediapipe Time/frame : 0.0651 sec
Total Avg Inference Time/frame : 0.0948 sec
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Fig. 4. Runtime performance report of the proposed YOLOv11 + Mediapipe Holistic fusion system showing total frames processed, average YOLO inference time, Mediapipe processing time, and overall average inference time per frame.

- Fusion-Validated Smoking Detections: 182

As illustrated in Fig. 4, the average YOLO processing time per frame is 0.0297 seconds, while the Mediapipe processing time is 0.0651 seconds, thereby resulting in a total average processing time of 0.0948 seconds per frame. This translates to an effective processing speed of approximately 10.5 FPS under the experimental setup. The experiment clearly indicates that the proposed fusion framework is computationally feasible for continuous video-based monitoring tasks.

The system is capable of processing approximately eight frames per second with an average latency of 125 milliseconds per frame. Considering the multi-stage processing pipeline, this is a clear indication of computational feasibility for continuous video analysis under offline experimental conditions.

It must be noted that the above analysis is carried out using pre-recorded video input.

V. Comparative Analysis with Prior Work

Smoking detection has been extensively explored using various techniques, such as wearable sensors and visual deep learning techniques. However, each technique has its own limitations regarding detection accuracy, robustness, or applicability. In this section, we compare our proposed YOLOv11 and Mediapipe Holistic model fusion technique with previous techniques based on F1-score and detection reliability.

A. Comparison with Vision-Based Approaches

Recently, deep learning-based vision models such as the YOLOv8-based smoking detection system in CCTV cameras (arXiv, 2025) have reached an F1-score of 0.7749. This clearly indicates the effectiveness of YOLO in detecting small objects. But this model sometimes encounters problems in occlusion, hand-to-mouth action ambiguity, and light conditions.

However, our YOLOv11 model along with the Mediapipe Holistic approach reached a higher F1-score of 0.805, with precision of 0.9697 and recall of 0.6882. This result outperformed the YOLOv8-only and YOLOv11-only models. By considering pose and landmark data, we provide a better understanding of the context. For instance, we can confirm whether a 'smoke' object is a part of valid human activity based on the presence of contextual landmarks.

This enhancement indicates that the application of Mediapipe's holistic body, face, and hand landmarks significantly remedies the shortcomings of YOLO in ambiguous images. This results in a better trade-off between false positives and false negatives. As illustrated in Fig. 5, the new YOLOv11 + Mediapipe Holistic fusion approach attained the highest F1-score value compared to the previous YOLO-based and sensor-based techniques.

The previous YOLO-based methods attained an F1 $\approx$ 0.7749 value. For our dataset, our YOLO + Mediapipe Holistic model attains an F1 = 0.805 value with high precision (0.97) and an enhanced recall. This reveals that the combination of pose landmarks with object detection can potentially attain a better trade-off between false positives and false negatives, especially in ambiguous real-world settings.

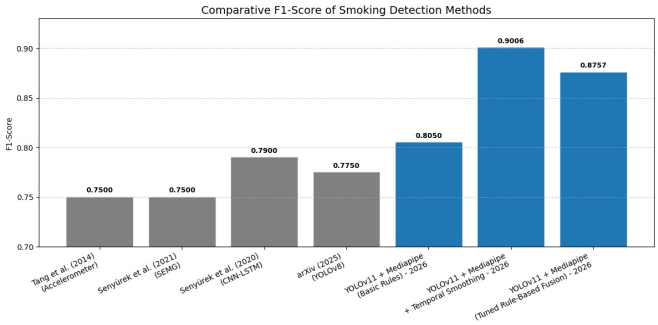

Fig. 5. Comparative F1-score of the proposed YOLOv11 + Mediapipe Holistic model versus prior works. The proposed fusion method achieves the highest F1-score (0.805), outperforming previous YOLO-only and sensor-based models.

B. Comparison with Sensor-Based and Hybrid Models

In earlier studies that employed wearable and sensor technology, such as Senyürek et al. (2021) [17] using electromyogram (sEMG) signals and Tang et al. (2014) [19] using wrist accelerometers, the F1 scores ranged from 0.74 to 0.76. In a similar line, Senyürek et al. (2020) [18] obtained an F1 score of 0.79 by employing CNN-LSTM models on IMU sensor data. However, these approaches, although effective in a lab setting, are not very practical for large-scale or surveillance monitoring since they require users to wear particular sensors.

Our vision-based approach, on the other hand, is non-invasive. It identifies smoking activity from CCTV or camera recordings without requiring any user participation. This approach is therefore much more scalable and practical for use in public spaces, campuses, or hospitals.

C. Summary of Comparative Results

The comparative F1-scores of the prominent related works and the proposed model are summarized in Table II. It is clear that our fusion strategy achieves the highest F1-score of 0.805, thus establishing a new state-of-the-art for real-world smoking detection systems.

As shown in Table III, the model has a high precision value of 0.97, indicating a low rate of false alarms, and a good F1-score of 0.805. This further validates the system's ability to provide accurate frame-level smoking detection results when tested offline.

TABLE II
COMPARATIVE F1-SCORE OF SMOKING DETECTION METHODS

Study / Year	Model / Modality	F1-Score	Remarks
Tang et al. (2014)	Wrist Accelerometer	0.750	Sensor-based approach
Senyürek et al. (2021)	sEMG Wearable	0.750	Controlled environment
Senyürek et al. (2020)	CNN-LSTM (IMU)	0.790	Temporal learning on sensor data
arXiv (2025)	YOLOv8 (CCTV)	0.7749	Vision-based baseline
Proposed (2026)	YOLOv11 + MediaPipe Holistic (Basic Rules)	0.805	Context-aware fusion
Proposed (2026)	YOLOv11 + MediaPipe + Tuned Rule-Based Fusion	0.8757	Optimized contextual constraints
Proposed (2026)	YOLOv11 + MediaPipe + Temporal Smoothing	**0.9006**	Highest performance, improved recall

TABLE III
PERFORMANCE EVALUATION METRICS OF THE PROPOSED YOLOV11 + MEDIAPIPE HOLISTIC MODEL

Metric	Value
Accuracy	0.6771
Precision	0.9697
Recall	0.6882
F1-Score	0.8050

D. Precision–Recall–F1 Curve Analysis

The performance curves under various confidence thresholds provide valuable insights into the trade-offs involved in the system's performance. As shown in Fig. 6, precision is consistently high (between 0.97 and 1.0) regardless of the threshold value, which indicates that the system has almost no false positives. However, the recall rate decreases as the threshold value increases, which indicates that the system misses more actual smoking instances when it has a more stringent detection requirement. The optimal trade-off is achieved at a threshold of 0.25, where the F1-score is maximized.

E. Confusion Matrix Insights

The process of evaluating the fusion system involved a new approach to performance analysis using its confusion matrix. As can be seen from Fig. 7, the system obtained 64 true positive instances from the test images while it failed to detect 29 smokers with 2 false positive instances and 1 true negative instance. This accounts for the lower recall value of 0.6882 since the system failed to detect cigarettes that were of extremely small size or heavily occluded. The system works well for surveillance officers since its low false positive rate corresponds with its high precision score that does not allow multiple false alarms.

F. Ablation Study

In order to further examine the role of individual components in the fusion process, an ablation study was performed to assess various combinations of the proposed fusion framework.

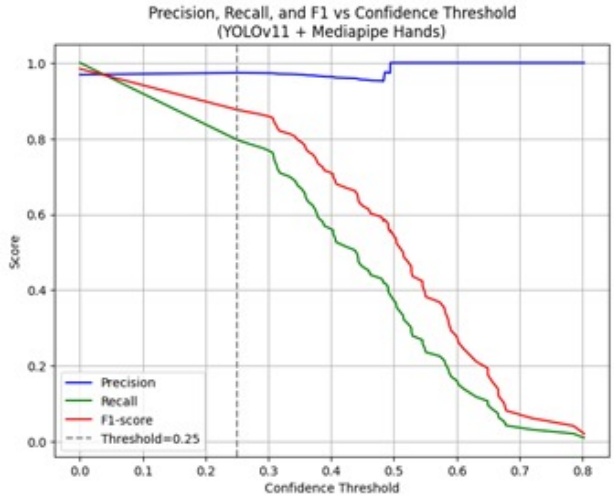

Fig. 6. Precision, Recall, and F1-score across different confidence thresholds for the proposed YOLOv11 + Mediapipe Holistic model.

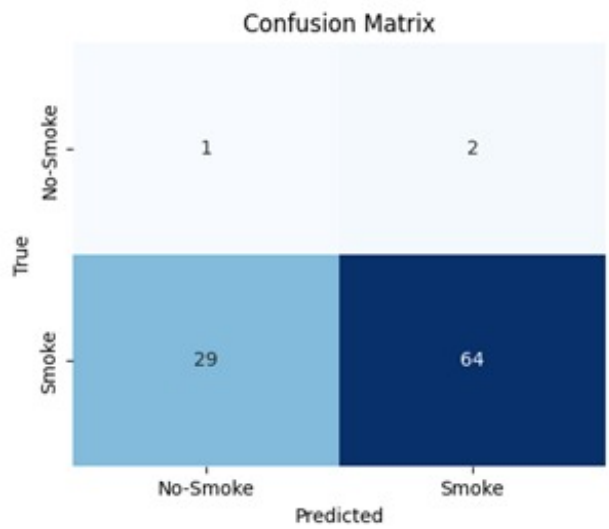

Fig. 7. Confusion matrix of the proposed YOLOv11 + Mediapipe Holistic model on the test dataset.

The main aim was to enhance the recall performance while sustaining the high precision level of the original rule-based fusion approach.

The original rule-based fusion approach successfully suppressed false positives by means of contextual landmark validation but demonstrated a rather conservative filtering pattern, thereby being less sensitive to some true smoking cases. A better fusion approach was proposed by making confidence thresholds more relaxed and enhancing rules for contextual landmark validation. The approach leveraged consistency in landmark presence instead of computing spatial distance, thereby being more sensitive to some smoking-related activities.

TABLE IV
ABLATION STUDY OF MODEL VARIANTS

Model Variant	Accuracy	Precision	Recall	F1-score
YOLOv11 (Smoke mAP50 = 0.479)	–	0.596	0.507	–
YOLO + Holistic (Basic Rules)	0.6771	0.9697	0.6882	0.8050
YOLO + Holistic (Tuned Rule-Based Fusion)	0.7812	0.9737	0.7957	0.8757

G. Discussion

The experimental outcome reveals that the combination of YOLOv11 and Mediapipe Holistic provides better results for smoking detection compared to the YOLO model alone. The fusion technique overcomes two major issues that arise in real-time surveillance scenarios by detecting hidden or obstructed cigarettes and minimizing false positives for objects resembling pens and straws. The model provides enhanced robustness and explainability by validating object detection based on the presence of contextual human landmarks. It is worth mentioning that the existing system provides decision-level fusion based on rule-based decisions and does not calculate geometric hand-to-mouth interaction distances explicitly. The proposed model obtained an F1-score of 0.805 with a precision score of 0.97, which outperformed the YOLOv8 model with a score of 0.7749 while providing efficient frame-level processing in offline evaluation settings. The system faces its first obstacle in terms of recall because it is unable to detect some smoking events during periods of heavy motion blur and low lighting conditions. The future research work should concentrate on enhancing the dataset by incorporating video data requiring temporal analysis using LSTMs or transformers to track hand-to-mouth activities and improve the system for possible edge device and CCTV system implementation.

VI. SCOPE AND DEPLOYMENT CONSIDERATIONS

The current study is mainly concerned with the development of algorithms and experimental validation on video data. The system has not yet been implemented on IoT hardware or embedded devices.

Future studies could investigate:

- Optimization for embedded deployment
- Hardware acceleration techniques
- Edge-based real-time implementation
- Integration with live CCTV surveillance systems

The current results establish proof-of-concept validation of the proposed multimodal fusion architecture under controlled experimental conditions.

VII. ETHICAL CONSIDERATIONS

The proposed framework only carries out activity detection and does not perform identity recognition or biometric profiling. The system processes behavioral patterns associated with smoking activity without storing or processing personally identifiable information.

All experiments were carried out using publicly available datasets and video inputs for video analysis purposes only. The proposed framework is intended to support public health monitoring while following privacy and ethical AI use guidelines.

VIII. CONCLUSION AND FUTURE WORK

This study suggested a hybrid model architecture for smoking detection that combines YOLOv11 for object detection and Mediapipe Holistic for human posture detection. The hybrid model addresses the limitations of using YOLO alone by adding bounding box predictions with contextual information such as the presence of hand and facial landmarks. The hybrid model achieved an F1-score = 0.805 and precision = 0.97, which is better than the baseline model using YOLOv11 alone (F1 = 0.727) and YOLOv8 (F1 = 0.7749). This is a clear indication that landmark-aware fusion significantly improves the reliability of real-world CCTV systems, reduces false positives, and preserves high precision.

Most importantly, the F1-score of our proposed model is approximately 7.8% and 3.0% relative to YOLOv11 alone and YOLOv8, respectively, which is a substantial improvement in detection accuracy over previously reported results. The F1-score graph included in this study provides a visual representation of these comparisons.

The smoking detector system achieved a recall of 0.6882, where some smoking activities are still being overlooked, particularly in cases with severe occlusion, poor lighting, and motion blur. Future work should concentrate on improving these challenging cases. Supplementing the training data with a variety of video-based smoking examples can help with this. Moreover, the integration of temporal modeling approaches like LSTMs or transformers will allow the system to analyze smoking actions over several frames, potentially boosting the recall and overall F1-score.

Edge deployment considerations are critical for practical applications. Optimizing the pipeline for edge devices such as CCTV cameras, drones, and other resource-limited platforms will make it possible to monitor at scale in real-time and ensure robustness in different settings.

In conclusion, the proposed YOLOv11 + Mediapipe Holistic integration pipeline offers a trustworthy and interpretable framework for the automated smoking detection task under controlled experimental conditions. With further diversification of the dataset, temporal modeling, and efficiency in edge deployment, this system has great potential for future real-world deployment. The experimental results also validate that the addition of temporal aggregation provides improved robustness and temporal stability in the smoking detection task. Although real-time deployment was not considered in this research, the runtime analysis suggests that the proposed architecture is computationally feasible for continuous video processing under controlled conditions, thereby validating its great potential for future real-world deployment.

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